NLP and Text Analysis: Introduction

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CBS

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Copenhagen Business School
HANDELSHØJSKOLEN

Natural Language Processing:
Processing of human language by computer

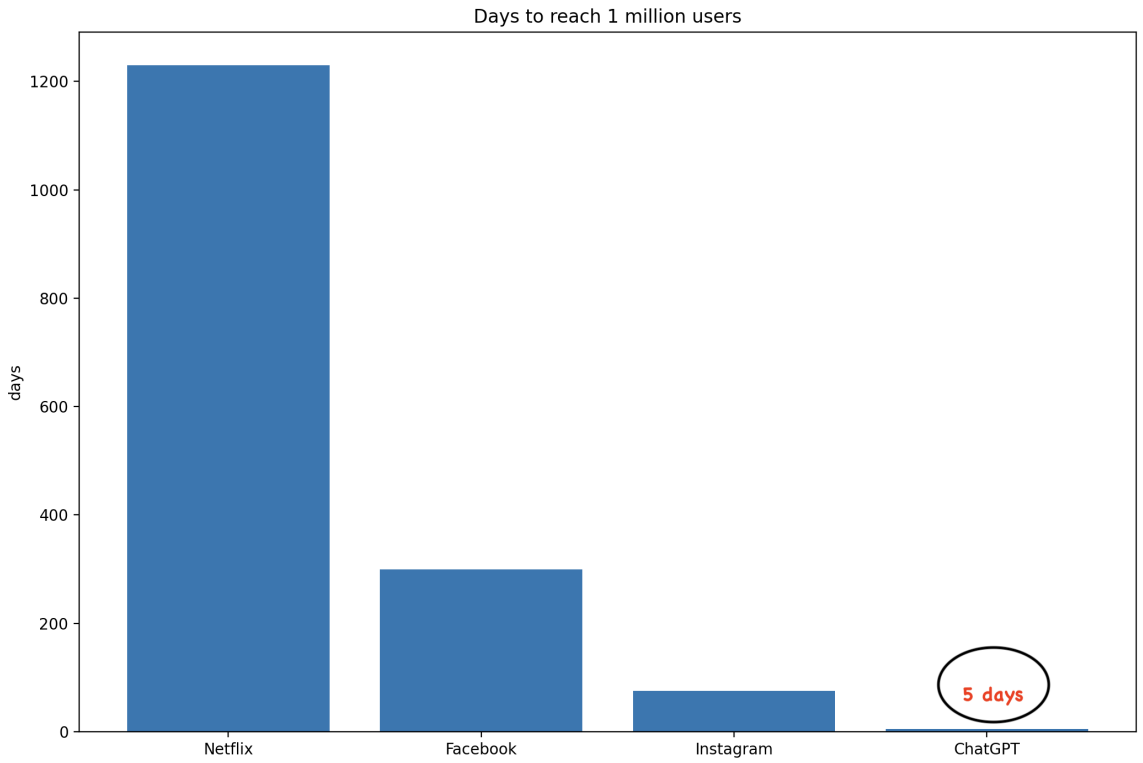
Tower of Babel

ChatGPT: the Technology

- Transformer model
- Pretraining on missing word prediction
- Multimodal – image and sound
- Reasoning models

ChatGPT

November 2022



The Endgame

Humanity May Achieve the Singularity Within the Next 3 Months, Scientists Suggest

Predictions across the field range from a few months to a few decades, but experts agree: change is coming.

BY DARREN ORF PUBLISHED: OCT 02, 2025 11:34 AM EDT

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Singularity

AGI

Superintelligence

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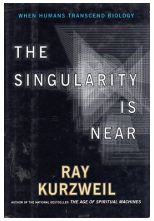
The Singularity

- AI will pass the Turing Test by 2029
- The Singularity will be reached by 2045 – “human life will be irreversibly transformed . . . future machines will be human-like, even if they are not biological”



Ray Kurzweil (2005)

AGI

Artificial General Intelligence:

"matches or surpasses human cognitive capabilities across a wide range of cognitive tasks"

AGI: Tests

- **The Robot College Student Test (Goertzel):** A machine enrolls in a university, taking and passing the same classes that humans would, and obtaining a degree. LLMs can now pass university degree-level exams without even attending the classes.
- **The Employment Test (Nilsson):** A machine performs an economically important job at least as well as humans in the same job. AIs are now replacing humans in many roles as varied as fast food and marketing.

AGI: Tests

- **The Ikea test (Marcus):** Also known as the Flat Pack Furniture Test. An AI views the parts and instructions of an Ikea flat-pack product, then controls a robot to assemble the furniture correctly.
- **The Coffee Test (Wozniak):** A machine is required to enter an average American home and figure out how to make coffee: find the coffee machine, find the coffee, add water, find a mug, and brew the coffee by pushing the proper buttons. This has not yet been completed.
- **The Modern Turing Test (Suleyman):** An AI model is given $100,000 and has to obtain $1 million

Machines of Loving Grace

Dario Amodei

Machines of Loving Grace[1]

How AI Could Transform the World for the Better

October 2024

I think and talk a lot about the risks of powerful AI. The company I'm the CEO of, Anthropic, does a lot of research on how to reduce these risks. Because of this, people sometimes draw the conclusion that I'm a pessimist or "doomer" who thinks AI will be mostly bad or dangerous. I don't think that at all. In fact, one of my main reasons for focusing on risks is that they're the only thing standing between us and what I see as a fundamentally positive future. **I think that most people are underestimating just how radical the upside of AI could be**, just as I think most people are underestimating how bad the risks could be.

Machines of Loving Grace

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Machines of Loving Grace[1]

How AI Could Transform the World for the Better

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Machines of Loving Grace

- "Powerful AI"
- Could come by 2026
- Smarter than a Nobel Prize winner across multiple fields
- All the interfaces – text, audio, video, internet access
- ...
- "a country of geniuses in a datacenter"

Language is Hard!

Current LLMs seem to have achieved human-level abilities to:

- process language
- reason using language

Is this really true?

- Many thinkers have argued that this could never happen

Descartes and AI

Could a machine imitate a human?

Rene Descartes: Discourse on the Method, Part V (1637)

Descartes and AI

Could a machine imitate a human?
No – you would always be able to tell the difference

Rene Descartes: Discourse on the Method, Part V (1637)

Descartes: Machines can't Imitate Humans

"...they could never use speech or other signs as we do when placing our thoughts on record for the benefit of others."

"...we can easily understand a machine's being constituted so that it can utter words, and even emit some responses to action on it ...But it never happens that it arranges its speech in various ways, in order to reply appropriately to everything that may be said in its presence, as even the lowest type of man can do.

Descartes: Machines can't Imitate Humans

" . . . although machines can perform certain things as well as or perhaps better than any of us . . . they infalliby fall short in others . . . For while reason is a universal instrument which can serve for all contingencies, these organs have need of some special adaptation for every particular action."

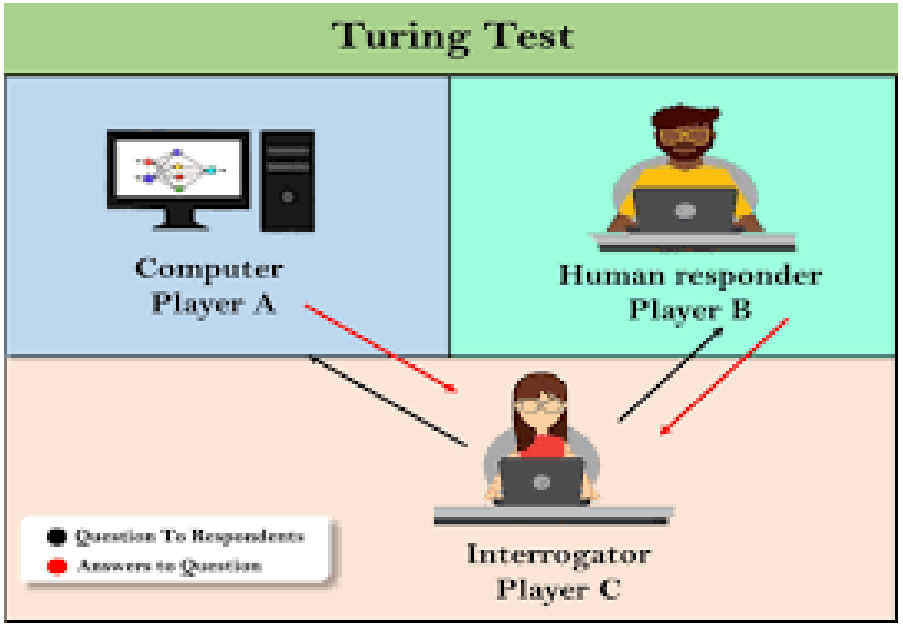
" . . . impossible that there should be sufficient diversity in any machine to allow it to act in all the events of life in the same way as our reason causes us to act."

The Turing Test

(1950) Computing Machinery and Intelligence

- Test of a machine's ability to exhibit intelligent behaviour
- human judge engages in a natural language conversation with a human and a machine
- If the judge cannot reliably tell the machine from the human, the machine passes the test

The Turing Test



Why is Language Hard?

Why is Language Hard?

Language is infinite

Why is Language Hard?

Language is infinite

Most sentences you hear –

Why is Language Hard?

Language is infinite

Most sentences you hear –

- you have never heard them before,

Why is Language Hard?

Language is infinite

Most sentences you hear –

- you have never heard them before,
- and will never hear them again

Why is Language Hard?

Language is ambiguous

Why is Language Hard?

Language is ambiguous

Many words have multiple meanings

Why is Language Hard?

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Many words have multiple meanings

Lexical Ambiguity

Why is Language Hard?

Language is ambiguous

Many words have multiple meanings

Lexical Ambiguity

Phrases and sentence can have multiple meanings

Why is Language Hard?

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Lexical Ambiguity

Phrases and sentence can have multiple meanings

Structural Ambiguity

Why is Language Hard?

Language is ambiguous

Many words have multiple meanings

Lexical Ambiguity

Phrases and sentence can have multiple meanings

Structural Ambiguity

A single sentence can have many different meanings

Why is Language Hard?

Language is ambiguous

Need Context to resolve ambiguities

Types of Data

- Numerical Data
- Categorical Data
- Text data is different
  - Content of an email
  - A Headline
  - Text of political speeches

Can text be treated as structured data?

Making Language into Structured Data

- The solution: Dummy values for words
  - One feature for each word
  - Value is 1 if word occurs in text, 0 otherwise
  - Alternative values: number of word occurrences, or TFIDF score
  - For any text, value of most features will be 0

Exploit Big Data for Text

- Use Supervised ML for text processing
- Can we get labeled text data?
- Build Classifiers
  - Spam Detection
  - Sentiment Analysis
  - Topic Detection
  - ...
- What does this have to do with AI? Real understanding? Turing Test?

Treat text as a **Bag of Words**

Bag of Words Processing

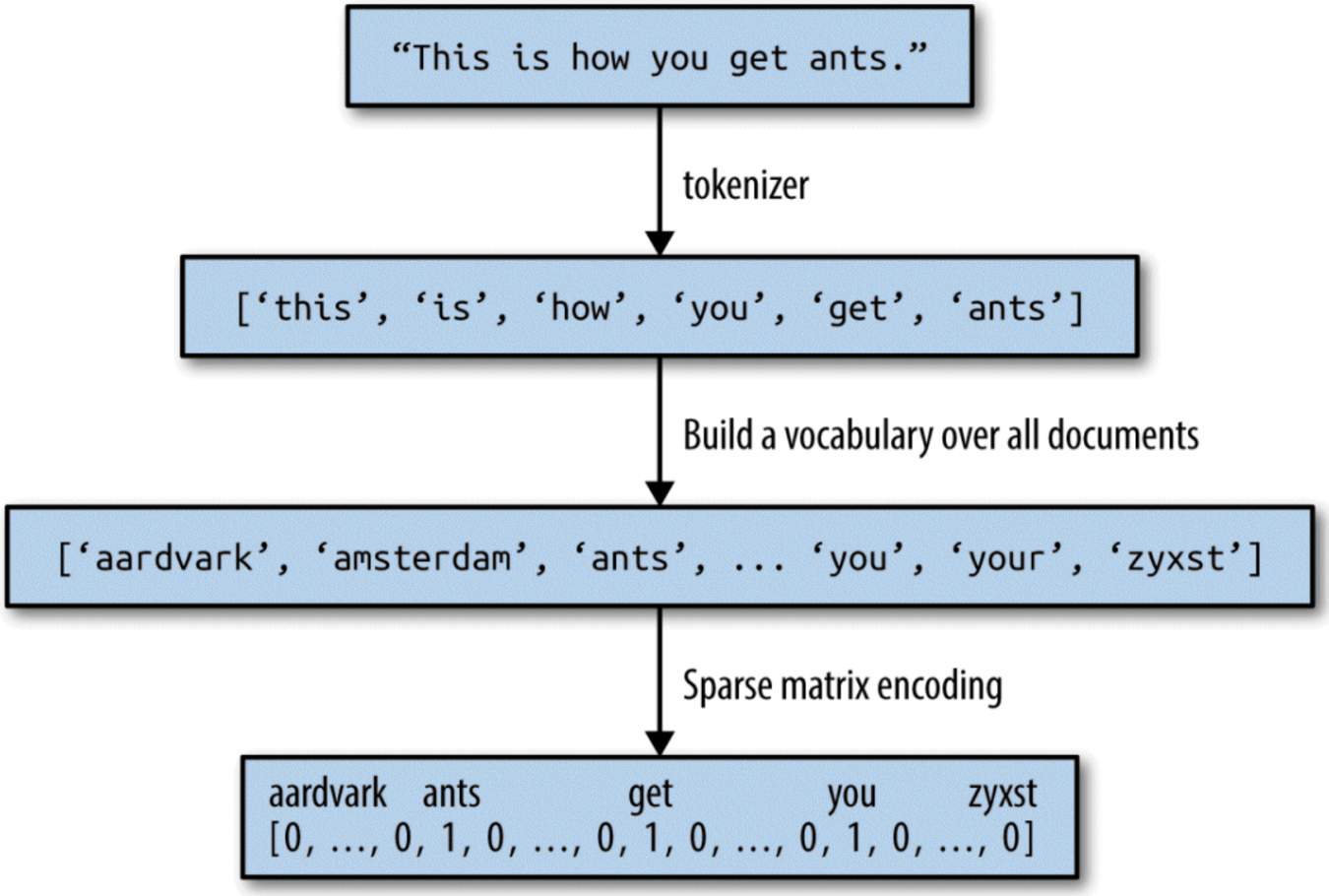


Figure 7-1. Bag-of-words processing

Markov Assumption

Andrei Markov

Simplifying assumption:

$$P(\text{the} \mid \text{its water is so transparent that}) \approx P(\text{the} \mid \text{that})$$

$$P(\text{the} \mid \text{its water is so transparent that}) \approx P(\text{the} \mid \text{transparent that})$$

$$P(w_1 w_2 \ldots w_n) \approx \prod_i P(w_i \mid w_{i-k} \ldots w_{i-1})$$

Approximate each component in the product

$$P(w_i \mid w_1 w_2 \ldots w_{i-1}) \approx P(w_i \mid w_{i-k} \ldots w_{i-1})$$

Language Modeling and Ngrams

Assign probability to a sequence of words

What is p(He went to the store)?

Language Modeling and Ngrams

He went to the store

- 1-grams (unigrams): He, went, to, the, store (5)
- 2-grams (bigrams): He went, went to, to the, the store (4)
- 3-grams (trigrams): He went to, went to the, to the store (3)
- 4-grams: He went to the, went to the store (2)
- 5-grams: He went to the store (1)

Language Modeling and Ngrams

Bigram approximation:

p(He went to the store) =
p(went|He) * p(to|went) *p(the|to) * p(store|the)

Trigram approximation:

p(He went to the store) =
p(to|He went) *p(the|went to) * p(store|to the)

Language Models

- "A general language model (LM) should be able to compute the probability of (and also generate) any string." (Radford et al. 2019)

Guess the Next Word

As Descartes said, I think, therefore I . . .
For all intents and . . .
I learned how to drive a . . .

Guess the Next Word

As Descartes said, I think, therefore I am
For all intents and . . .
I learned how to drive a . . .

Guess the Next Word

As Descartes said, I think, therefore I am
For all intents and purposes
I learned how to drive a . . .

Guess the Next Word

As Descartes said, I think, therefore I am
For all intents and purposes
I learned how to drive a car

Guess the Next Word

Monica: Okay, everybody relax. This is not even a . . .
Rachel: Oh God... well, it started about a half hour before the . . .
Ross: (squatting and reading the instructions) I'm supposed to attach a brackety . . .

Guess the Next Word

Monica: Okay, everybody relax. This is not even a date
Rachel: Oh God... well, it started about a half hour before the
...
Ross: (squatting and reading the instructions) I'm supposed to attach a brackety ...

Guess the Next Word

Monica: Okay, everybody relax. This is not even a date
Rachel: Oh God... well, it started about a half hour before the wedding
Ross: (squatting and reading the instructions) I'm supposed to attach a brackety . . .

Guess the Next Word

Monica: Okay, everybody relax. This is not even a date
Rachel: Oh God... well, it started about a half hour before the wedding
Ross: (squatting and reading the instructions) I'm supposed to attach a brackety thing

The Training: Next Word Prediction

This is a pointless task

It's also impossible

But it's also genius

The Training: Next Word Prediction

- This is Supervised Machine Learning – each example is *labeled* with correct answer
- But no human labeling is needed – we just take text from the Internet, and hide a word
- Essentially unlimited amount of data

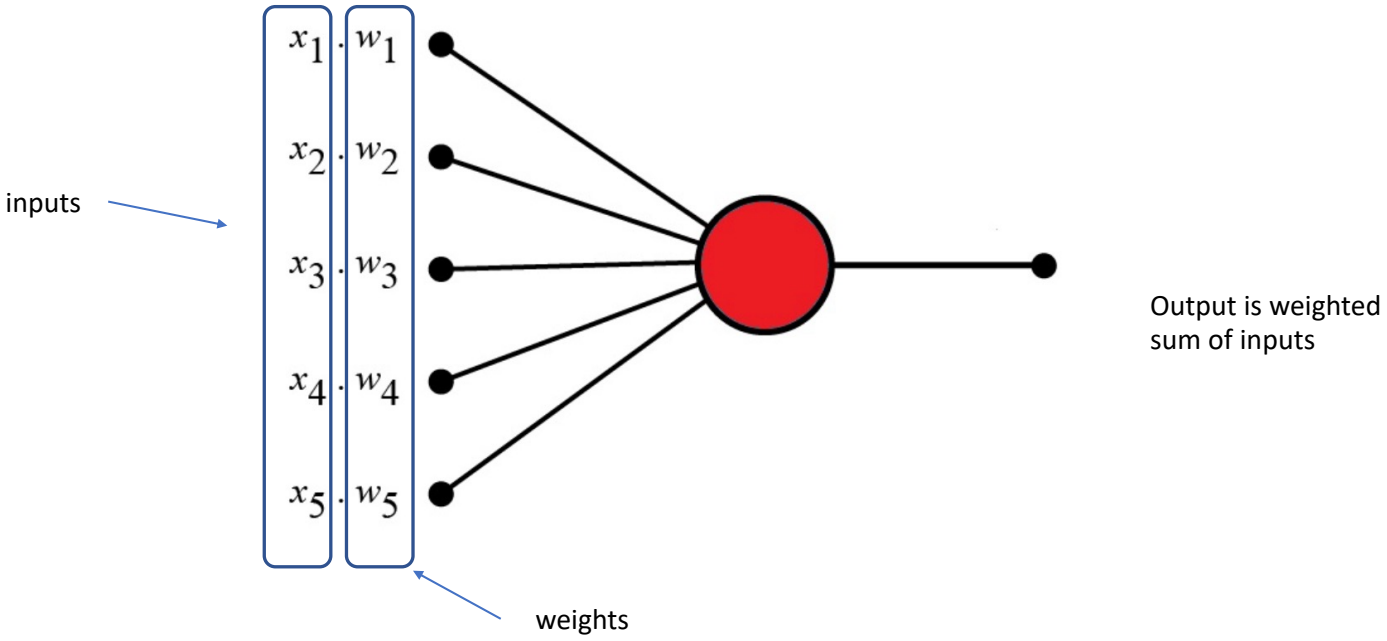
The Training: Next Word Prediction

System outputs a probability distribution for each word in vocabulary:

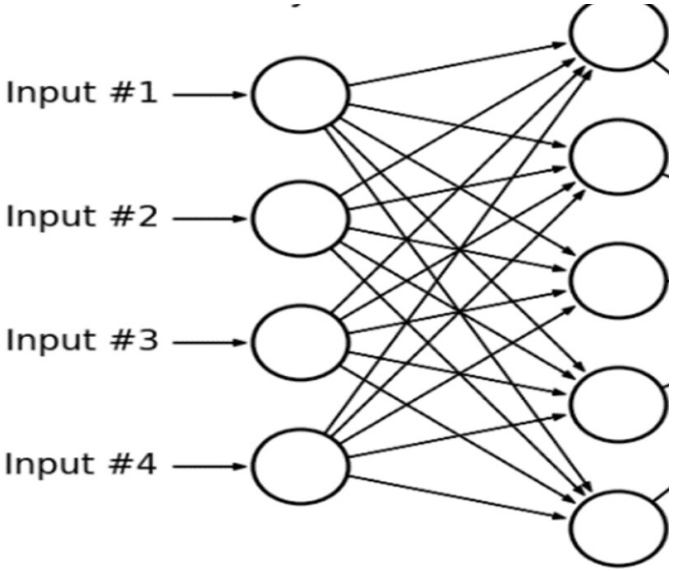
Thanks for all the [fish/.05, help/.04, support/.01, …]

This talk is about [chatGPT/.02,AI/.019,business/.017, …]

A Node of a Neural Network



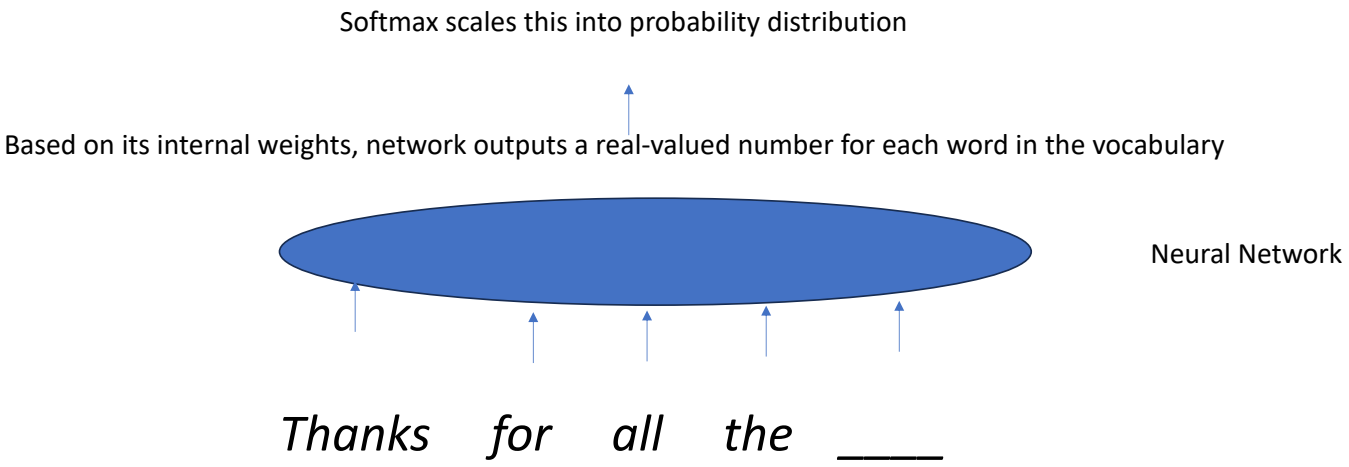
A Neural Network for Word Prediction



Output value for every word in vocabulary

fish	12,567.2
help	123
the	-3,198
aardvarks	-22
table	-1107

The Training: Next Word Prediction



The Training: Next Word Prediction

- Let's say this is first sentence in training – so maybe all words get the same probability
- With, say 50,000 words in vocabulary, we would have output like this:

[fish/.00002, help/.00002, ..., the/.00002, ...aardvarks/.00002]

The Training: Next Word Prediction

- We compute a **loss function**
- Masked word was “fish”
- Loss = 1 - prob(“fish”)
- We want to minimize loss

- So if model said prob(“fish”) = 1, loss would be 0
- If model said prob(“fish”) = 0, loss would be 1

Best possible

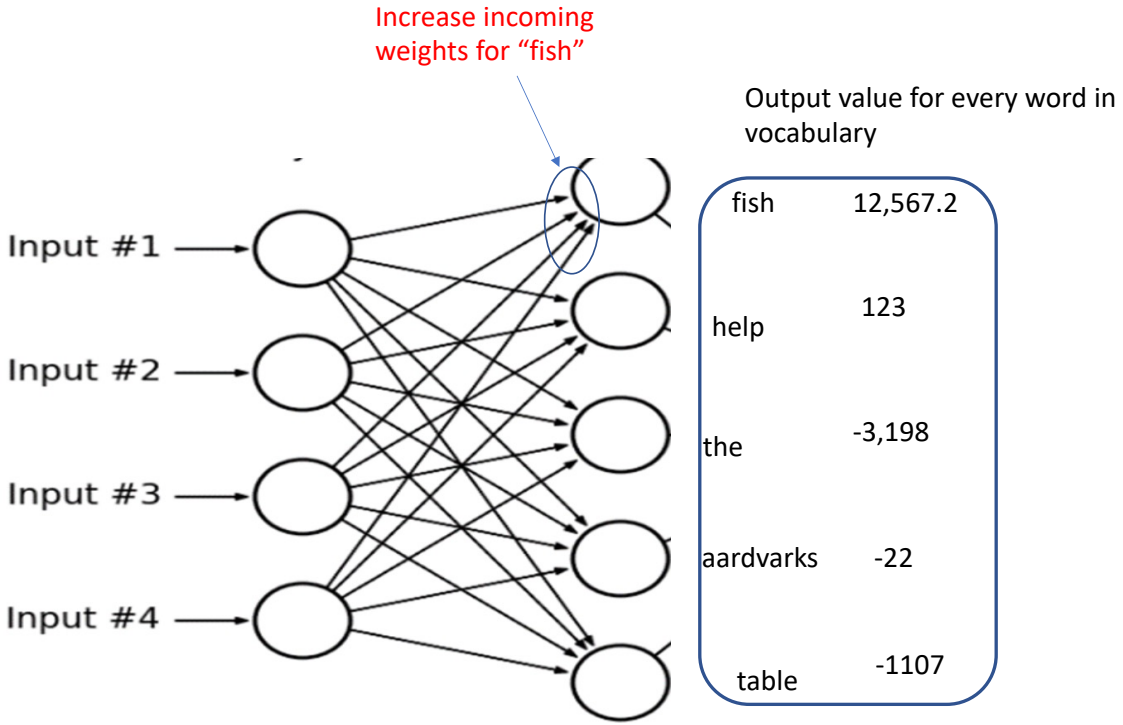
Worst possible

The Training: Next Word Prediction

Now, learning happens!

- Modify network weights to increase value computed for “fish”

A Neural Network for Word Prediction



Next Word Prediction: Learning

For this example:

- Decreases loss
- Increases probability for “fish”
- *Also, it will slightly reduce the probability for every other word in vocabulary*

The Training: Next Word Prediction

Each example produces a very small amount of useful information

- Make “fish” slightly more likely in the given context
- Also -- make every other word slightly less likely in this context

We repeat this an insane amount of times

Language Ability of Current LLMs

- LLMs (Large Language Models) today generate highly coherent, grammatical texts that can be indistinguishable from human output
- They exhibit at least some knowledge of hierarchical structure and abstract linguistic categories

(Mahowald et al. 2024)

Language Ability of Current LLMs

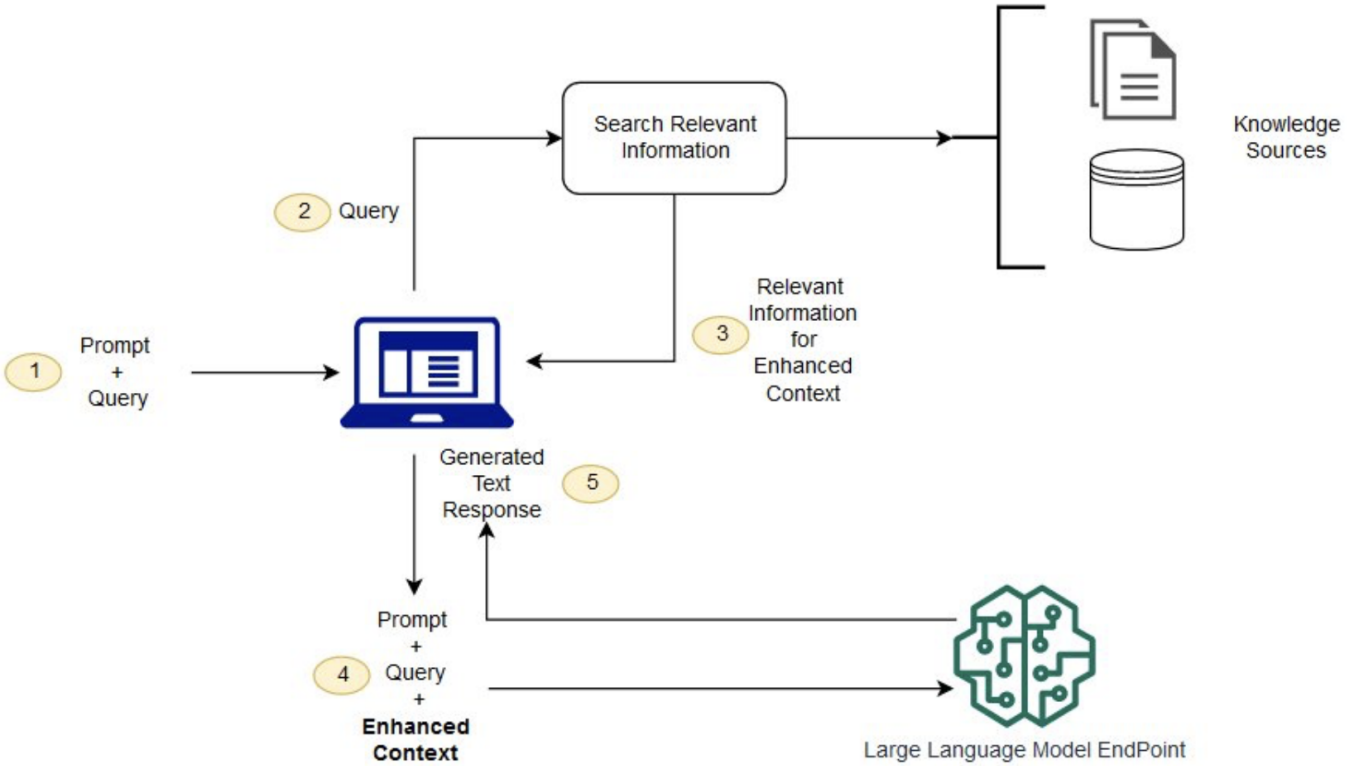
- These models are not perfect learners of abstract linguistic rules, but neither are humans
- LLMs are on track to acquiring formal linguistic competence
- LLMs have already overturned the claims about the fundamental impossibility of acquiring certain linguistic knowledge—including hierarchical structure and abstract categories—from the input alone

(Mahowald et al. 2024)

LLM Techniques

- Prompt Engineering
- Few-shot learning
- RAG

RAG


Search Relevant Information
Knowledge Sources
2 Query
1 Prompt + Query
3 Relevant Information for Enhanced Context
Generated Text Response
5
4 Prompt + Query + Enhanced Context
Large Language Model EndPoint

Application Examples

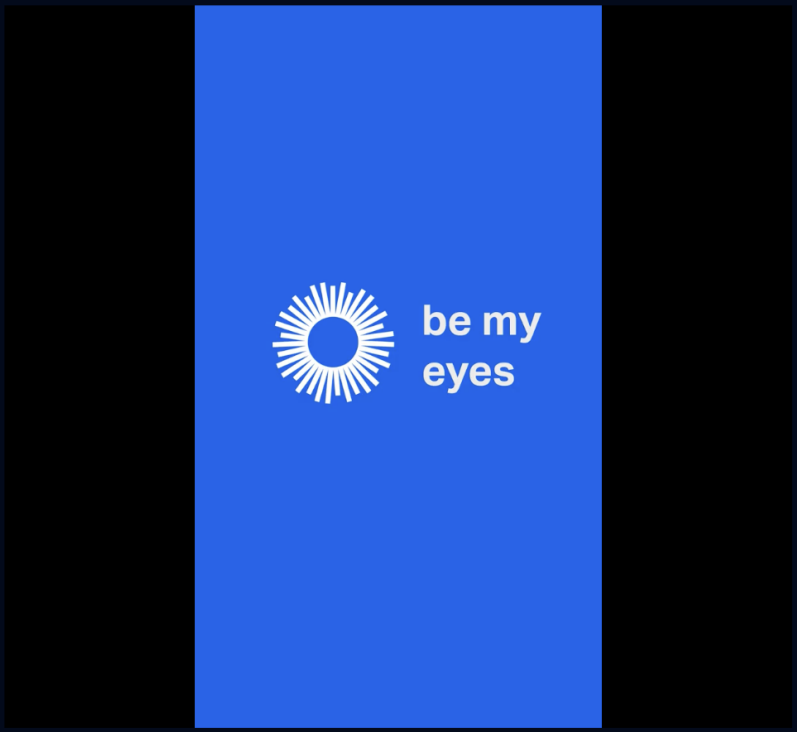
- RAG interface for
  - customer support
  - onboarding
- CBS Research portal with RAG
- BeMyAI

BeMyAI

Now available

Be My AI™
AI-powered visual assistance.

Be My AI™ integrates with Microsoft



Blind YouTube vlogger and TV presenter Lucy Edwards introduces Be My AI™.

Takeaways

- Language is Hard!
  - Language is infinite
  - The challenge of ambiguity – both lexical and structural

Takeaways

- The Revolution in NLP
- Current LLM's
  - Appear to have (close to) human-level language ability
  - Exciting research directions!
    - Building applications exploiting LLM's
    - Probing their abilities
  - Powerful AI (Machines of Loving Grace) is coming soon